

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T26 · Solving Inequalities

Questions

Student book · questions with writing space

■ 23 pages

FAMILY MAP

Solving Inequalities

F01	Solve linear inequalities	10 Q 27 marks
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F02	Solve quadratic inequalities	5 Q 17 marks
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F03	Translate inequality notation and number lines	7 Q 27 marks
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9 (a) Solve the inequality $7 - 3t < 2t + 15$

(2)

- 19** Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

Question 21

4MA1-2H Q15

MODULAR UNIT 2

3 marks

- (b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

(3)

Question 04

4MA1-1H Q8 (a)

MODULAR UNIT 2

2 marks

8 (a) Solve the inequality $8x - 4 \geq 3x - 10$

.....

(2)

Question 07

4MA1-1H Q8 M03

MODULAR UNIT 2

3 marks

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

.....
(3)

Question 33

4MA1-2H Q20 M01

MODULAR UNIT 2

3 marks

- 20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

.....

Question 28

4MA1-2H Q15 Q15(b)

MODULAR UNIT 2

3 marks

- (b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

(3)

Question 06

4MA1-1H Q6 Q6(a)

MODULAR UNIT 2

2 marks

6 (a) Solve the inequality $5x - 7 \leq 2$

(2)

Question 34

4MA1-2H Q22

MODULAR UNIT 2

5 marks

22 Here is a rectangle.

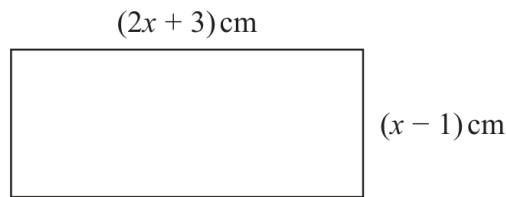


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

7 (a) Solve the inequality $2x + 7 > 4$

(2)

(2)

A horizontal number line with arrows at both ends. It has tick marks at every integer from -5 to 5, labeled with their respective numbers: -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5. The line is labeled with a t at the right end.

(a) Write down all the possible values of y .

Show your working clearly.

(3)

A horizontal number line with arrows at both ends. It has tick marks at every integer from -6 to 6. The numbers are labeled below the line: -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6. The line is labeled 'x' at the right end.

(3)

A horizontal number line with arrows at both ends. It has tick marks at every integer from -6 to 6. The numbers are labeled below the line: -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6. The line is labeled 'x' at the right end.

1 (a) Simplify $g^6 \times g^4$

(1)

(b) Simplify $k^{10} \div k^3$

(1)

(c) Simplify $(3cd^4)^2$

(2)

(d) Solve the inequality $4x + 7 > 2$

(2)

(Total for Question 1 is 6 marks)

(2)

(2)

Show clear algebraic working.

A horizontal line segment with arrows at both ends, labeled $(2x - 4) \text{ cm}$.

$(3x + 2)$ cm

Given that $A < 3x + 27$
find the range of possible values for x .

(2)

A number line graph showing the inequality $x \leq 3$. The number line is marked from -2 to 5. A solid black dot is placed at the number 3. An arrow points to the left from the dot, indicating that the solution set includes all real numbers less than or equal to 3.

(1)

(c) Solve the inequality $7w + 6 > 12w + 14$

- 22** Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

.....
(Total for Question 22 is 3 marks)

- 25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)